

MACE HEAD, Ireland

WMO: 039630

Lat: 53.3267N

Lon: 9.9043W

Elev: 75

StdP: 14.66

Time Zone: 1.00 (IRL)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.5	33.3	23.5	17.8	32.9	26.8	20.8	35.4	49.2	44.8	45.2	44.7	10.7	50	0.825

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	69.1	62.1	66.1	60.2	63.9	59.2	63.3	67.3	61.7	64.7	60.6	62.9	11.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.7	82.4	64.8	60.5	78.9	62.9	59.5	76.1	62.0	28.7	67.3	27.7	64.9	26.8	62.9	68.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
41.8	36.7	32.8	DB	28.8	75.2	3.9	4.8	26.0	78.7	23.8	81.5	21.6	84.2	18.8	87.7
			WB	26.9	65.6	4.1	2.7	23.9	67.5	21.5	69.1	19.2	70.6	16.2	72.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	44.1	43.7	45.2	48.8	52.6	56.7	59.2	59.0	57.0	52.8	47.8	45.4
	DBStd	6.81	4.35	4.03	4.16	3.60	3.99	3.62	3.07	2.10	2.81	3.56	4.27	4.95
	HDD50	842	184	177	154	63	12	0	0	0	0	15	88	149
	HDD65	5099	649	596	614	487	384	253	185	188	240	378	517	608
	CDD50	1230	1	1	5	26	94	201	285	278	210	102	21	6
	CDD65	9	0	0	0	0	1	3	4	1	0	0	0	0
	CDH74	25	0	0	0	0	4	14	6	1	0	0	0	0
	CDH80	1	0	0	0	0	0	1	0	0	0	0	0	0
Wind	WSAvg	16.7	19.0	18.4	16.2	15.0	15.4	13.7	13.7	16.1	16.9	17.6	18.4	20.5
Precipitation	PrecAvg	65.6	7.0	5.6	4.6	4.0	3.9	4.0	4.4	5.1	5.2	6.5	7.4	7.4
	PrecMax	75.9	12.0	10.7	8.6	8.9	7.3	7.0	6.7	11.4	9.5	12.6	13.3	13.0
	PrecMin	50.9	2.4	1.6	2.3	1.8	1.1	2.0	2.2	1.4	0.9	2.2	3.6	3.1
	PrecStd	7.0	2.3	2.4	1.5	1.5	1.7	1.4	1.3	2.3	2.3	2.6	2.2	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.6	53.1	59.4	64.1	71.8	75.4	73.9	68.9	67.4	63.1	57.5	54.8
		MCWB	51.1	50.1	50.9	54.9	62.1	63.2	65.9	62.0	60.7	59.5	55.7	52.9
	2%	DB	51.2	51.3	54.0	58.6	65.3	68.3	68.9	65.1	64.2	60.2	55.9	53.6
		MCWB	49.8	49.5	48.8	51.4	57.7	60.4	63.5	60.2	59.7	57.9	54.2	52.2
	5%	DB	50.4	50.2	51.4	55.7	61.7	64.7	65.7	63.3	62.0	58.5	54.7	52.4
		MCWB	49.1	48.9	48.3	50.7	55.5	58.5	61.5	59.1	59.0	56.0	52.8	51.0
	10%	DB	49.8	49.3	50.2	53.8	58.5	62.0	63.6	62.1	60.8	57.5	53.7	51.4
		MCWB	48.4	47.9	47.8	49.5	53.5	57.2	59.8	58.4	57.8	55.2	51.7	49.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.8	51.0	51.9	55.8	62.2	64.2	66.6	63.5	62.4	60.5	56.4	53.9
		MCDB	52.3	51.9	56.3	62.8	70.9	73.1	72.5	67.6	64.1	61.8	56.8	54.2
	2%	WB	50.6	50.1	50.3	52.9	58.7	61.3	64.1	61.6	61.1	58.8	54.8	52.6
		MCDB	51.0	50.9	52.3	56.0	64.6	67.6	67.8	63.6	62.8	60.1	55.7	53.2
	5%	WB	49.8	49.3	49.4	51.6	56.4	59.3	62.4	60.7	60.0	57.0	53.5	51.3
		MCDB	50.3	50.2	51.0	54.3	60.4	63.3	65.1	62.3	61.6	57.9	54.6	52.2
	10%	WB	48.9	48.4	48.4	50.6	54.7	58.0	60.8	59.8	58.9	56.0	52.2	50.5
		MCDB	49.8	49.4	50.1	53.4	57.6	61.1	62.9	61.6	60.5	57.2	53.6	51.6
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.8	7.8	9.1	8.8	8.2	7.0	6.5	6.7	6.9	6.6	6.1
		MCDBR	5.5	5.8	8.9	11.7	14.4	13.3	10.0	7.9	8.2	7.1	6.3	6.0
		MCWBR	6.4	5.8	6.7	7.4	8.6	7.6	6.4	5.7	6.3	6.5	6.8	6.7
	5% WB	MCDBR	5.0	5.6	7.8	9.6	12.4	10.7	9.0	6.8	6.8	5.7	6.5	5.9
		MCWBR	6.2	5.9	6.7	6.9	7.8	6.7	6.3	5.8	6.3	6.2	7.1	6.8
Clear-Sky Solar Irradiance	taub	0.337	0.344	0.359	0.376	0.385	0.368	0.373	0.361	0.361	0.343	0.332	0.326	
	taud	2.276	2.365	2.380	2.356	2.373	2.458	2.453	2.498	2.481	2.471	2.329	2.243	
	Ebn at Noon	195	239	261	271	274	280	276	274	259	238	197	173	
	Edh at Noon	20	25	30	35	36	33	33	30	27	22	19	17	
All-Sky Solar Radiation	RadAvg	211	429	799	1282	1595	1674	1485	1265	915	533	259	155	
	RadStd	20	38	69	106	122	145	108	97	68	42	22	19	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page